

Recent changes in heavy metal contamination at near-channel positions of the Lahn River, central Germany

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ABSTRACT

Short-term monitoring of heavy metal concentrations along rivers affords the opportunity to examine the persistence of contaminants in the fluvial environment. This paper examines changes in the concentrations of Cu, Pb, and Zn at near-channel positions along a 75-km reach of the Lahn River, an agrarian watershed in central Germany, between 2000 and 2007. Thirty-three sites sampled in 2000 were re-sampled in 2007 in order to document changes in trace metal contamination. The concentrations of Cu and Pb decreased between 2000 and 2007, whereas the concentration of Zn was statistically unchanged. The Pb concentration decreased at 82% of the sample sites over the study period, whereas Cu concentrations decreased at 70% of the sites. Site-to-site variation in the concentrations of Pb with distance downstream from potential source areas declined between 2000 and 2007. The downstream variation in Cu also decreased over the study period, albeit to a lesser degree. The decrease in Pb, and to a lesser extent Cu, concentrations adjacent to the Lahn River appears to reflect the recent reductions in the release of trace elements to the environment. Zinc concentrations were unchanged, suggesting that it is derived from different sources than are Cu and Pb or is cycling through the watershed differently than Cu and Pb. Lower concentrations of Cu and Pb in channel bank sediments translate to a reduced risk of future downstream heavy metal pollution in the study area.

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1. Introduction

Surface waters in Europe have long been exposed to substantial levels of anthropogenic contaminants such as heavy metals, in some cases dating back several thousands of years (Brännvall et al., 1999; Bindler et al., 2011). The emission of metals to the environment is generally agreed to have peaked in the first half of the twentieth century and has been decreasing for the last several decades (Middelkoop, 2002). With declining releases to the environment, the concentrations of heavy metals in river systems of Western Europe began to decrease in the latter half of the twentieth century (Vink et al., 1999; Middelkoop, 2002). Middelkoop (2002) noted a rapid decline in metal pollution along the lower Rhine River in the Netherlands between 1975 and 1985, followed by a slower rate of decline after 1985. With the end of the Cold War, the spread of pollutants by both air and water from Eastern Europe was further reduced as many aged industrial plants were closed and modern wastewater treatment facilities were installed (Scherer et al., 2003; Zerling et al., 2006).

In German watersheds, heavy metals are among the most studied pollutants. Usually the product of industrial or mining activity, they have been discharged to the environment for centuries (Matschullat et al., 1997; Brännvall et al., 1999; Renberg et al., 2001), with the

greatest releases occurring from the mid-1800s until the early 1970s (Valette-Silver, 1993; Middelkoop, 2002). Although surface water systems in Germany have become markedly cleaner over the last few decades, high concentrations of heavy metals remain along rivers, especially in industrial and mined areas of the country. The August 2002 floods along the Elbe River of northern Germany served notice that metals are present in high concentrations across floodplains and can be remobilized and spread downstream during high water events (Förstner, 2003; Krüger et al., 2005; Schwartz et al., 2006).

Given the greater potential impact on human population and the likelihood of high metal concentrations, most research in Germany and surrounding countries has focused on urbanized, industrial, or mined watersheds (e.g., Middelkoop, 2002; Förstner, 2003; Macklin et al., 2006; Schwartz et al., 2006; Wijnhoven et al., 2006; Bábek et al., 2008; Ciszewski and Turner, 2009; Dennis et al., 2009). This body of work documents anthropogenic heavy metal contamination along German rivers such as the Rhine and Elbe. Since the mid-1990s, I have examined the storage and distribution of heavy metals in the Lahn River basin (Martin, 1997, 2004, 2009), a relatively agrarian watershed draining ~5000 km² in west-central Germany (Fig. 1). My work demonstrates that the aggregate storage of heavy metals (Cu, Pb, Zn) along the Lahn River is less than that noted along waterways such as the Rhine or Elbe, but individual sites have metal concentrations approaching levels normally found in urbanized watersheds (Martin, 2004). Heavy metal concentrations are at a maximum in near-channel sediments (<100 m from active channel) and at depths of 15–20 cm; at depths >25 cm,

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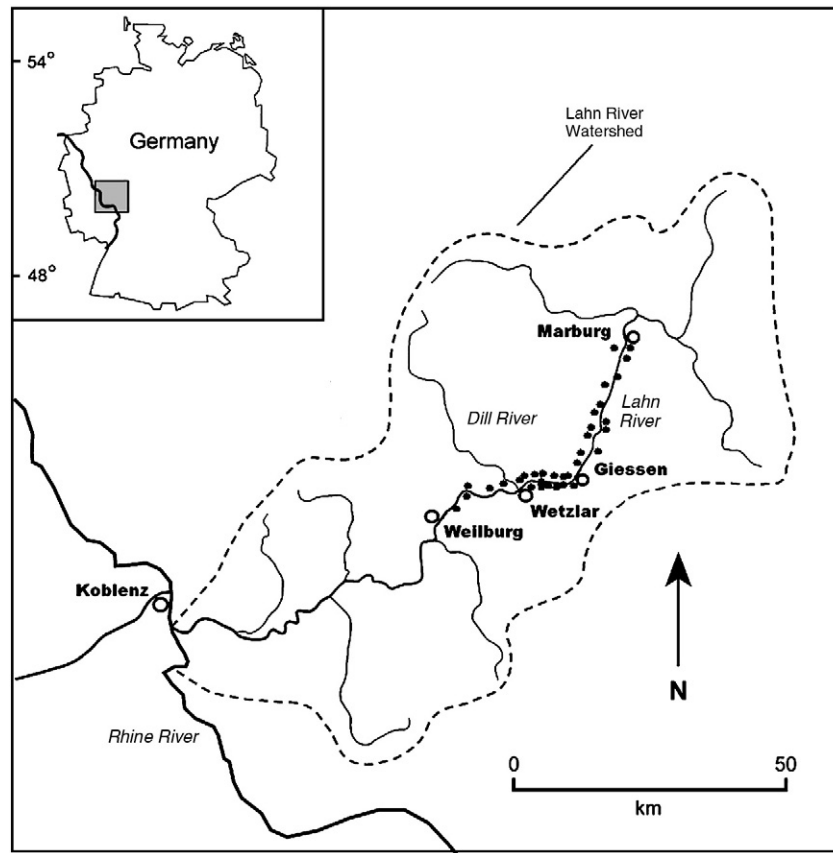


Fig. 1. Study reach of the Lahn River between Marburg and Weilburg. Sample locations ($n = 33$) are shown with solid dots.

the concentrations are virtually identical to concentrations in deeply buried sediments (Martin, 1997, 2000). Heavy metal concentrations are generally higher in the vicinity of and downstream from urban areas and industrial facilities, especially in the valley reach between Giessen and Weilburg (Martin, 2004); these “hot spots,” or sites with higher metal concentrations, appear similar to those reported along urbanized and mined watersheds in England (Coulthard and Macklin, 2003). The surface samples at most sites along the Lahn River contain lower concentrations of heavy metals than the samples collected at 15–20 cm, likely reflecting the recent decline in metal releases to the environment, a trend noted in other studies (Brüggmann, 1995; Middelkoop, 2002; Zerling et al., 2006).

The present study was undertaken to assess the changes in heavy metal concentrations since 2000 at channel bank locations along a 75-km reach of the Lahn River between Marburg and Weilburg. Channel banks where I collected samples in 2000 were resampled in 2007. Additionally, their locations were recorded with a handheld GPS unit to allow future monitoring of changes in metal concentrations. The objectives of the study are to (i) determine short-term changes in heavy metal concentrations in the upper 5 cm of floodplain soils at near-channel (<5 m from Lahn River) locations, (ii) look for changes in the downstream pattern of heavy metal concentrations since 2000, and (iii) evaluate the effect of organic matter content and grain size on heavy metal concentrations and determine whether changes occurred since 2000. This study affords a detailed look at short-term changes in the concentrations of heavy metals in channel bank sediments of the Lahn River. Channel bank positions are relatively active geomorphically and have therefore been identified as potentially significant sources for heavy metal pollution (James, 1989; Marcus, 1989; Leigh, 1997; Leece and Pavlowsky, 2001). In addition to documenting short-term change in metal concentrations, the study also provides an opportunity to examine the short-term cycling of metals through the Lahn River watershed.

2. Study area

The Lahn River, with a total drainage area of ca. 5000 km² at its mouth south of Koblenz, has its headwaters in the rural uplands north of Marburg (Fig. 1). Between Marburg and Giessen, the river flows southward through a broad valley before turning westward at Giessen toward Wetzlar and Weilburg. The basin is underlain by shales and greywacke of Paleozoic age as well as Tertiary basalt and sandstone (Knoblich and Sanner, 1992). Black, organic-rich silt was deposited in the Lahn River and surrounding valleys during the late Holocene (generally the last 3000 years) to a thickness of ca. 3.0 m along the Lahn River south of Marburg (Bos and Urz, 2003) and 2.3 m in the Amöneburger basin to the northeast of Giessen (Rittweger, 2000). Farther south, Houben (2003) reported 1–2 m of massive late Holocene alluvium underlying floodplains of the Wetter River east of Giessen. The Lahn River floodplain along much of the 75-km study reach is broad, reaching a maximum width of 2.5 km just downriver of Giessen. Land use in the Lahn River drainage basin is 70% agriculture (crops and grassland), 16% forested, and 14% urban (Jöckel, 1981). The watershed between Giessen and Wetzlar is more urbanized than the other parts of the study reach. Away from the urban centers of Marburg, Giessen, and Wetzlar, the floodplain in the study reach is cultivated or grazed; above the floodplain, uplands are largely forested. No physical evidence or archival record has been found of large-scale mining in the Lahn River watershed between Marburg and Wetzlar; but the Dill River, which joins the Lahn at Wetzlar, flows through a region where iron ore was mined from the late nineteenth century until the early 1980s.

Similar to many rivers in Germany, the Lahn was deepened and channelized in the mid-nineteenth century. Several small step-dams were built in the study reach in subsequent decades, reducing flow velocity and stream power. Overbank flow is most common between January and April, when precipitation tends to increase and early spring

snowmelt in upland areas contributes to higher flows. Flooding has the capacity to locally erode the channel and deposit fine sands on point bars and near-channel surfaces of the floodplain. The examination of aerial photographs taken between 1935 and 1995, together with considerable field experience in the Lahn River basin since 1995, indicates that much of the channel has been stable for some time. The presence of meander scars in the study reach suggests, however, that earlier lateral migration was common (Martin, 2000). Floodplain sediments consist of black, organic-rich silt thought to have accumulated during the late Holocene (Mäkel, 1969; Houben, 2003) when surrounding hillslopes were cleared of forest cover for agriculture and timber (Rittweger, 2000; Kalis et al., 2003; Rommens et al., 2006; Hoffmann et al., 2007). Surface soils have a mean pH of 5.2 (standard deviation 0.5) and a clayey silt texture. Channel bank riparian vegetation consists of scattered deciduous trees and a ca. 3-m-wide, near-channel vegetation zone composed of low-lying brush. Cultivation, where present, generally does not extend to the channel bank.

3. Methods

I used personal field notes, topographic maps, and ground-based photographs to relocate 33 sites where I collected channel bank samples in 2000. Sample sites were positioned <5 m from the active river channel. Sites with anthropogenic levees were avoided because the date of levee construction could not be determined; topographic maps distinguish between natural and anthropogenic levees. In addition, I avoided reaches that were close to major roads, rail lines, and industrial facilities and sampled narrow and wide-valley reaches. Access to the river channel was limited in some locations, but I attempted to collect samples from an equal number of left and right channel bank locations (Fig. 1).

At each site, a core 2.5 cm in diameter was extracted from the floodplain surface using a stainless-steel, Oakfield handheld corer. To allow future monitoring of heavy metal concentrations, the latitude and longitude of each sample site was recorded using a handheld GPS unit. Samples were transported in sterile plastic bags to the geomorphology lab at Justus Liebig-Universität in Giessen, where the upper 5 cm of each core was sampled. Samples were then split, with half retained for standard sediment analyses and half submitted to ALS Chemex Labs (an ISO 9002 registered analytical laboratory in North America) for analysis of heavy metals. To ensure comparability of samples collected in the 2000 and 2007, samples were submitted to the same ALS Chemex lab facility in Vancouver, BC, Canada, used in previous work and identical laboratory preparation procedures (air drying of samples, screening to 63 μm) were followed. Heavy metal concentrations were determined from the <63 μm fraction. Prior to analysis of heavy metals, samples were digested in perchloric, nitric, and hydrofluoric acids (triple-acid digestion), a process that yields near total recovery of metals (Leece and Pavlowsky, 1997). Inductively coupled plasma atomic emission spectroscopy (ICP-AMS) was used to measure concentrations of Cu, Pb, and Zn; concentrations of Cd were also measured, but the results are omitted here because the analytical capabilities for Cd (0.5 mg kg^{-1}) differ little from the concentrations found in surface soils (Martin, 2000, 2004). The analytical sensitivity was 1.0 mg kg^{-1} for Cu and 2.0 mg kg^{-1} for Pb and Zn. Quality control data supplied by ALS Chemex revealed coefficients of variation of 1% for Cu, 5% for Pb, and 3.5% for Zn on repeated measurements of standards. Analysis of duplicates produced a mean difference of 1.25 mg kg^{-1} for Cu, 0.75 mg kg^{-1} for Pb, and 5.8 mg kg^{-1} for Zn. On the sample half that was retained, I determined the proportions of clay (<2 μm) and silt (2–63 μm) using the pipette method; sand percentage was determined through wet sieving. Organic matter content was measured by loss on ignition (LOI) at 430 °C for 24 h (Davies, 1974).

In order to evaluate the magnitude of heavy metal enrichment in surface soils, I compared surface concentrations to those measured in background samples. Background samples, which are presumed to

represent the pre-industrial age (pre-1750 A.D.) metal concentrations, were collected at depths of 0.85 to 1.15 m along the study reach of the Lahn River. Of the 35 background samples, 18 were collected at channel bank sites where surface samples were collected; at the other 17 sites where surface samples were collected, I was unable to core to depths necessary to collect background samples because of sand and gravel layers or a high water table. The other 17 background samples came from locations within the study reach, but at sample sites <10 m from the Lahn River channel. Background samples were analyzed by ALS Chemex (Vancouver facility) with the same procedures used for surface samples. Ascertaining background, or nonanthropogenic, metal concentrations is problematic (Birch et al., 2000; Matschullat et al., 2000; Bindler et al., 2011). I opted to use deeply buried sediments to acquire background metal concentrations based on previous work along the Lahn River (Martin, 2000). In addition, I have AMS conventional radiocarbon ages of 300 ± 40 RCYBP (Beta-201178) and 210 ± 40 RCYBP (Beta-201179) on charcoal recovered from 1.05 and 0.90 m, respectively, at two separate sites near the mouth of the Dill River, a tributary to the Lahn River. These ages suggest that sediments found at a depth of ca. 1 m in the study area are pre- or early-industrial in age and contain nonanthropogenic concentrations of metals.

The German Federal Soil Protection Act of 1998 (*Bundes-Bodenschutzgesetz* {BbodSchG}) and the German Federal Soil Protection and Contaminated Sites Ordinance of 1999 (*Bundes-Bodenschutzverordnung* {BbodSchV}) have set the standards for various contaminants and transfer paths. The natural soil functions are said to be at risk when a 'precautionary value' is exceeded. The precautionary values for metals used here are for soil type loam/silt and are reported in mg kg^{-1} dry weight fine soil. Metal concentrations from the 2000 and 2007 samples were compared to the precautionary values to provide a measure of soil metal contamination and assess changes in metal enrichment between the two time periods. Threshold effect concentration (TEC) and probable effect concentration (PEC) guidelines developed by MacDonald et al. (2000) were used as an additional measure of metal contamination and change in metal enrichment. The TEC represents the level below which adverse impacts are unlikely, whereas the PEC is the concentration above which negative effects are considered likely.

4. Results

4.1. Mean metal concentrations, 2000 and 2007

Between 2000 and 2007, the near-channel concentrations of Cu and Pb declined, whereas that of Zn increased (Table 1). Based on a paired *t*-test comparison of mean values, the declines in Cu and Pb concentrations are statistically significant ($p=0.05$), whereas the increase in Zn concentration is not. The maximum values for all three metals were less in 2007 than in 2000, with the decrease most dramatic for Pb (108 mg kg^{-1} in 2000 to 65 mg kg^{-1} in 2007). In 2000, nearly two-thirds of the sample sites (21 of 33) exceeded or equaled the precautionary value of 40 mg kg^{-1} for Cu; by 2007, only 14 sites exceeded or equaled the value. No sample sites in 2000 or 2007 had Cu concentrations that exceeded the PEC value (concentration above which adverse effects to the environment are likely). Eight sites in 2000 had Pb concentrations that exceeded or equaled the precautionary value of 70 mg kg^{-1} ; by 2007, no site exceeded or equaled the value. Five sample sites in 2000 had Pb concentrations that exceeded the PEC value; by 2007, no sample site had Pb concentrations exceeding the PEC. For Zn, the precautionary value of 150 mg kg^{-1} was exceeded or equaled at 32 sites in 2000 and in 2007. The PEC for Zn was exceeded at six sample sites in 2000, but at only one sample site in 2007. When examined on a sample site basis, the Cu concentration decreased at 23 sites (70%) between 2000 and 2007, increased at 9 sites (27%), and remained unchanged at 1 site (3%). At 27 sites (82%) the Pb concentration declined between 2000 and 2007; it increased at the other 6 sites

Table 1

Summary descriptive statistics for soil samples; the mean concentrations shown in bold are statistically different ($p = 0.05$); standard deviations (SD) are shown in parentheses; the "Precautionary Value" is published in the German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz*) of 1998; the TEC (threshold effect concentration) and PEC (probable effect concentration) values are from MacDonald et al. (2000); all values are for dry weight sediment.

	Metal concentrations (mg kg^{-1})		
	Cu	Pb	Zn
2000 ($n = 33$)			
Mean (SD)	45.4 (11.4)	62.2 (18.3)	240.9 (70.4)
Range	29–69	38–108	118–386
2007 ($n = 33$)			
Mean (SD)	41.1 (8.9)	48.5 (9.9)	242.9 (57.3)
Range	27–56	34–65	146–343
Precautionary			
Value	40.0	70.0	150.0
TEC	35.7	35.0	123.0
PEC	197.0	91.3	315.0
Background ($n = 35$)			
Mean (SD)	22.9 (12.7)	26.6 (23.3)	84.1 (43.8)
Range	11–82	12–136	42–246

(18%). The Zn concentration in 2007 was higher at 16 sites (48%), lower at 16 sites (48%), and unchanged at 1 site (3%).

4.2. Downstream metal concentration patterns

Martin (2004) demonstrated that near-channel metal concentrations change little along the Lahn River between Marburg and Giessen, but then increase downstream of Giessen to Wetzlar and farther downstream to Weilburg. When the 2007 data are included, this general downstream pattern for Cu, Pb, and Zn is still evident (Fig. 2). Based on downstream trends in metal concentrations, the study reach of the Lahn was divided into three river segments: an upstream segment between Marburg and Giessen, a middle segment from Giessen to Wetzlar, and a downstream segment between Wetzlar and Weilburg. The Marburg–Giessen segment had the lowest mean concentrations of Cu, Pb, and Zn in 2000 and in 2007 (Table 2). Along this segment, the mean concentrations of Cu and Pb were statistically lower in 2007 than in 2000, whereas the mean concentration of Zn was statistically the same in the two time periods. This segment of the river is the most agrarian and least impacted by urbanization and industrial activity. In the Giessen to Wetzlar segment, mean metal concentrations were higher than in the segment immediately upstream in 2000 and in 2007. The mean concentration of Pb was statistically lower in 2007 than it was in 2000. In contrast, the mean concentration of Zn was statistically higher in 2007 than in 2000; the mean concentration of Cu was statistically the same in the two time periods. The Wetzlar to Weilburg segment had the highest mean concentrations of Cu, Pb, and Zn in 2000 and in 2007. The mean concentrations of Cu, Pb, and Zn in this segment were statistically lower in 2007 than they were in 2000.

One possible source of the higher metal concentrations downstream of Wetzlar is metals introduced to the river in the industrial cities of Giessen and Wetzlar. A second source could be the Dill River, which drains an area that was extensively mined for iron from the late nineteenth century until the early 1980s. The Dill River empties into the Lahn River at Wetzlar. Samples ($n = 15$) collected in 2010 at channel bank locations along the Dill River upstream of Wetzlar reveal mean Cu concentrations of 73.8 mg kg^{-1} , mean Pb concentrations of 89.2 mg kg^{-1} , and mean Zn concentrations of 339.1 mg kg^{-1} , values that are higher than those noted along the Lahn River.

Although the mean concentration of metals is higher in the downstream segment (Wetzlar–Weilburg) than in the upstream segment

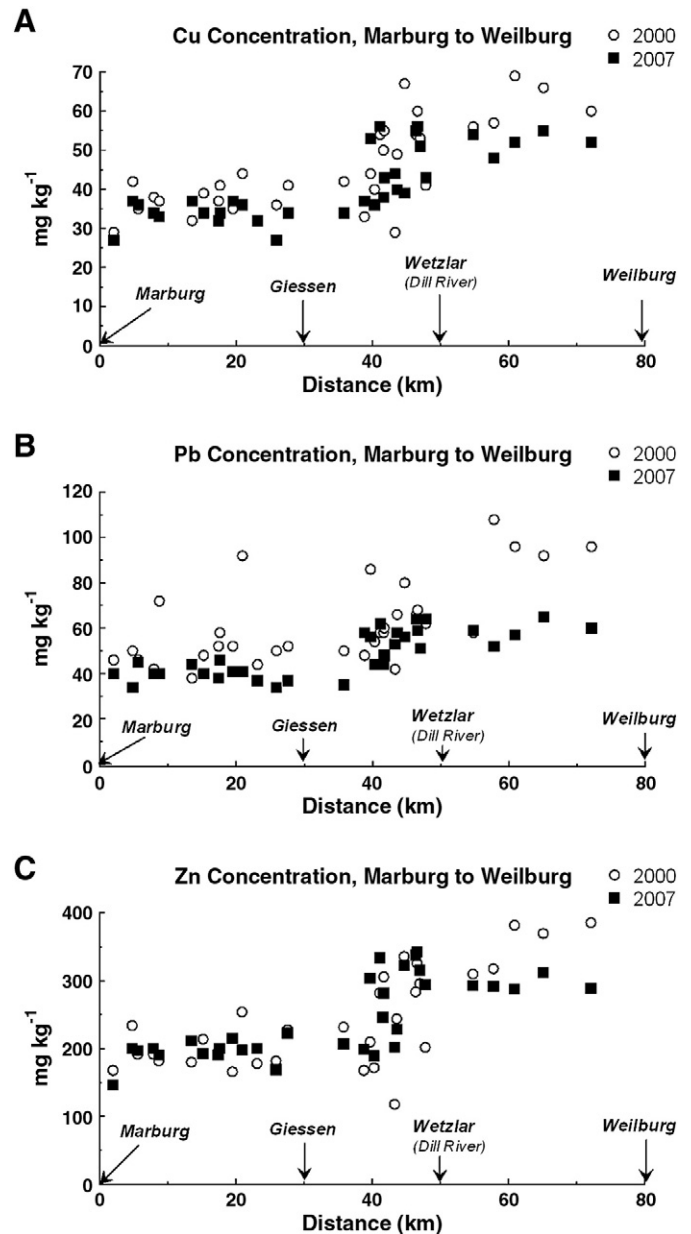


Fig. 2. Downstream changes in Cu, Pb, and Zn concentrations, 2000 and 2007.

(Marburg–Giessen), the increase in metal concentrations with distance downstream is not smooth but rather displays significant site-to-site variability, especially downstream of Giessen (Fig. 2). In contrast, site-to-site variability in metal concentrations is minimal between Marburg and Giessen. In comparing the 2007 trends to those of 2000, site-to-site variability in Cu and Pb concentrations appears to have decreased; little change in the site-to-site variability for Zn was noted over the time period. In 2000, the site-to-site variability for the three metals was greatest in the Giessen to Wetzlar river section; by 2007, this variability had decreased for Pb, and to a lesser extent for Cu, but was unchanged for Zn. The tendency toward a more uniform downstream trend in the concentrations of Cu and Pb suggests that sediments deposited in the near-channel environment since 2000 contained less Cu and Pb than did sediments that accumulated prior to 2000. Similar to the mean values depicted in Table 1, the downstream trends indicate that near-channel concentrations of Cu and Pb at individual sites were lower in 2007 than in 2000 (Fig. 2).

Table 2

Summary descriptive statistics for soil samples by river segment; for a given metal and river segment, mean concentrations shown in bold are statistically different ($p = 0.05$) between 2000 and 2007; standard deviations are shown in parentheses; river segments are Marburg–Giessen (Mb–Gi), Giessen–Wetzlar (Gi–Wz), and Wetzlar–Weilburg (Wz–Wb) (Figs. 1 and 2 show the river segment location).

	Metal concentrations (mg kg ⁻¹)		
	Cu	Pb	Zn
2000			
Mb–Gi ($n = 14$)	37.0 (4.2)	53.0 (13.8)	197.4 (26.0)
Gi–Wz ($n = 14$)	47.9 (10.4)	61.6 (5.7)	244.4 (64.9)
Wz–Wb ($n = 5$)	61.6 (11.8)	90.0 (18.9)	353.2 (36.4)
2007			
Mb–Gi ($n = 14$)	33.6 (3.3)	39.8 (3.6)	195.4 (19.0)
Gi–Wz ($n = 14$)	44.6 (7.9)	53.7 (8.5)	271.9 (57.7)
Wz–Wb ($n = 5$)	52.2 (2.7)	58.6 (4.7)	294.8 (9.8)
Background ($n = 35$)			
Mean (SD)	22.9 (12.7)	26.6 (23.3)	84.1 (43.8)
Range	11–82	12–136	42–246

4.3. Impact of sediment texture and organic content

Heavy metal concentrations are often associated with organic-rich, fine-textured sediments (e.g., de Groot et al., 1982; Horowitz, 1985; Bubb and Lester, 1996), although some studies report no correlation between metal concentrations and organic matter or fine sediment contents (Lokas et al., 2010). Near-channel alluvium along the Lahn River typically has a clayey-silt texture, with clay contents around 20% (Martin, 2009). The organic matter content of surface soils is typically 10–11% (Martin, 2009). In previous work along the Lahn River, Martin (2004) found virtually no relationship between the percentage of fine sediments and heavy metal content, but did discern a weak relationship between organic matter content and heavy metal concentrations.

Here, the relationship between organic matter content and metal concentrations and sediment texture and metal concentrations is stronger in 2007 than in 2000 (Table 3). The 2007 data reveal that increasing organic matter content and silt and clay content have a weak impact on the concentrations of Cu and Zn. In contrast, the clay content has a statistically significant, albeit weak, impact on Cu, Pb, and Zn concentrations. Organic matter content and silt and clay content have a statistically significant impact on Zn concentration in 2007, but they are statistically unrelated to Pb concentration.

5. Discussion

Building upon previous work (Martin, 2004), resampling of near-channel sites reveals elevated heavy metal concentrations relative to background levels in floodplain soils of the Lahn River, but indicates that cleaner sediments have recently been deposited in near-channel locations. For the entire study reach, the mean concentrations of Cu and Pb at near-channel sites declined between 2000 and 2007; in contrast, the mean concentrations of Zn were statistically unchanged over the time period (Table 1). When the study reach is divided into three river segments (Marburg–Giessen, Giessen–Wetzlar, Wetzlar–Weilburg), concentrations of Pb declined in all three segments between 2000 and 2007. In contrast, concentrations of Cu declined in only the Marburg–Giessen and Wetzlar–Weilburg segments between 2000 and 2007. Changes in Zn concentrations were more variable, increasing in the Giessen–Wetzlar segment but decreasing between Wetzlar and Weilburg from 2000 to 2007. Concentrations of Cu in the Giessen–Wetzlar segment and concentrations of Zn in the Marburg–Giessen section were statistically the same in 2000 and 2007.

Among the three heavy metals for the entire study reach, the concentration of Pb experienced the greatest decline. In 2000, eight near-channel sites had Pb levels that exceeded the precautionary value set

Table 3

Relationship between metal concentrations and percent organic matter (OM), percent clay (Cl), and percent silt plus clay (SiCl).

	<i>n</i>	<i>r</i> ²	<i>P</i>
2000			
Cu = 39.8 + 0.33 OM	33	0.08	0.11
Cu = 40.9 + 0.29 Cl	33	0.01	0.64
Cu = 40.7 + 0.08 SiCl	33	0.01	0.54
Pb = 52.1 + 0.59 OM	33	0.09	0.07
Pb = 55.3 + 0.46 Cl	33	0.01	0.65
Pb = 56.3 + 0.10 SiCl	33	0.01	0.63
Zn = 192.6 + 2.84 OM	33	0.15	0.02
Zn = 244.8 – 0.25 Cl	33	0.00	0.95
Zn = 228.8 + 0.21 SiCl	33	0.00	0.79
2007			
Cu = 30.3 + 1.11 OM	33	0.10	0.08
Cu = 17.9 + 1.39 Cl	33	0.25	0.00
Cu = 22.6 + 0.29 SiCl	33	0.18	0.01
Pb = 44.9 + 0.37 OM	33	0.01	0.60
Pb = 30.3 + 1.09 Cl	33	0.13	0.04
Pb = 33.0 + 0.25 SiCl	33	0.10	0.07
Zn = 162.9 + 8.2 OM	33	0.13	0.04
Zn = 99.9 + 8.6 Cl	33	0.23	0.00
Zn = 139.6 + 1.68 SiCl	33	0.14	0.03

forth in German federal soil pollution standards and five sites had Pb concentrations that were greater than the PEC value; by 2007, no site had a concentration that exceeded the German federal standard or the PEC value. The decline in Pb storage was widespread in the basin, with 82% of the sample sites recording a lower Pb concentration in 2007 than in 2000. The trend between 2000 and 2007 suggests that the decrease in Pb releases to the environment, which began in Western Europe in the 1970s, is finally reflected in the Pb movement through and storage in the Lahn River basin. Studies elsewhere in Western Europe have noted that metal concentrations in floodplain surface soils are frequently less than concentrations present at depths below the floodplain surface (e.g., Brüggmann, 1995; Gocht et al., 2001; Scherer et al., 2003; Zerling et al., 2006). Sediments deposited on the near-channel floodplain post-2000 contain less Pb than those that accumulated prior to 2000 (Martin, 2004). The five sample sites with the highest Pb concentrations in 2000 had decidedly lower concentrations of the metal in 2007 and no longer stood out as locations with anomalously high Pb concentrations (Fig. 2).

The concentration of Cu in the study area also decreased between 2000 and 2007, but the decline was less than that documented for Pb. The Cu concentration declined at 70% of the sample sites between 2000 and 2007; by 2007 only 14 sites exceeded the precautionary value for Cu, down from 21 sites in 2000. Sediments being stored in near-channel positions contain lower concentrations of Cu than those that accumulated before 2000. Similar to Pb, the storage of Cu in the study reach is beginning to reflect the aforementioned decrease in the release of heavy metals to the environment, mirroring trends documented along other rivers of Western Europe (e.g., Brüggmann, 1995; Gocht et al., 2001).

In contrast to Cu and Pb, the Zn concentration in sediments deposited in near-channel locations for the entire study reach was statistically unchanged between 2000 and 2007. When the study reach is divided into the aforementioned river segments, Zn concentrations were statistically unchanged between 2000 and 2007 in the Marburg–Giessen segment, statistically higher in 2007 than in 2000 for the Giessen–Wetzlar segment, and statistically less in 2007 than in 2000 for the Wetzlar–Weilburg segment (Table 2). Enrichment factors illustrate the degree of anthropogenic metal contamination in the study reach (Table 4). For the entire study reach of the Lahn River, the anthropogenic enrichment of Cu and Pb decreased between 2000 and 2007, whereas the enrichment of Zn was unchanged. Enrichment factors for Cu and Pb decreased at each of the three river segments from 2000 to 2007, but

changes by river segment for Zn were not uniform. The more complicated temporal changes in Zn concentration (i.e., increase in one river section, decrease in another) may result from Zn being a more soluble heavy metal (Gocht et al., 2001) and one with numerous anthropogenic sources (Charlesworth and Lees, 1999; Tobin et al., 2000) in contrast to Cu or Pb. This would also account for the higher enrichment factors for Zn relative to Cu and Pb in 2000 and 2007. Based on the persistence of relatively high concentrations of Zn in near-channel sediments between 2000 and 2007, the concentrations of Zn found in near-channel sediments in 2007 apparently were out-of-phase with the previously mentioned reductions in Zn releases to the environment.

The enrichment factor for Pb decreased more between 2000 and 2007 than did the enrichment factors for Cu and Zn, declining from 2.3 in 2000 to 1.8 in 2007; in contrast, Cu declined from 2.0 in 2000 to 1.8 in 2007 and Zn was 2.9 in both periods (Table 4). The relatively large decline in Pb concentration may have resulted from a decline in the inputs of Pb to the study reach from multiple and geographically widespread nonpoint sources. As indicated by several studies (e.g., Brännvall et al., 1999; Renberg et al., 2001; Middelkoop, 2002), Pb emissions peaked throughout Western Europe sometime between the late 1960s and mid-1970s, then began to drop steadily once unleaded gasoline was introduced in 1974 (De Vleeschouwer et al., 2007). In the former West Germany, the transition to unleaded fuel was dramatic. Unleaded fuel increased from 1% of the total automobile fuel sold in West Germany in 1984 to 89% by 1993; as a result, Pb deposition in bulk precipitation declined by an average of nearly 80% in four German states from 1984 to 1993 (Schulte et al., 1996). During the decades prior to the mid-1980s, when the burning of leaded fuel was prevalent, Pb likely accumulated in surface soils over a geographically broad area of the Lahn River watershed. That Pb is now being conveyed through and ultimately out of the Lahn River basin, a cycling that may take some time given how geographically dispersed the metal is through the watershed. The lower concentrations of Pb in 2007 compared to 2000 suggest that sediments with lower concentrations of Pb have recently begun to move through the Lahn River and are being deposited on channel banks. This has lowered the Pb concentration in surface soils and reduced downstream site-to-site variability since 2000.

A consideration of the downstream pattern of heavy metal concentrations also reveals changes in the storage and movement of contaminants between 2000 and 2007. In 2007, as was the case in 2000, the concentrations of Cu, Pb, and Zn generally increase in the downstream direction, likely a reflection of the more industrialized character of the watershed moving from Marburg to Giessen and Wetzlar and the addition of metal-contaminated sediments by the Dill River in Wetzlar. Metal concentrations in 2000 were fairly uniform between Marburg and Giessen, then increased downstream toward Weilburg (Martin, 2004). Site-to-site variability in metal concentrations reached a maximum between Giessen and Wetzlar. The overall pattern of increasing

concentrations with distance downstream was apparent in 2007 as it had been in 2000, but the site-to-site variability was less in 2007 than in 2000 (Fig. 2). This is especially true for Pb, which displayed far less variability between Giessen and Wetzlar in 2007 than was evident in 2000. The maximum values for each of the three metals were also less in 2007 than they had been in 2000, although decreases appear less for Zn than for the other two metals. The data suggest that sediments with lower concentration of Cu and Pb were deposited at the 33 sample sites along the river between 2000 and 2007, with the greatest decrease occurring in the Wetzlar–Weilburg section of the river (Table 2).

The downstream concentration of Cu decreased slightly between 2000 and 2007, although not to the extent that Pb concentration decreased over the same time period. One possible explanation for the difference in the concentration of the two metals is that the amount of Cu delivered to the river declined less between 2000 and 2007 than did the amount of Pb. Anthropogenic Pb was apparently more widespread and present in higher amounts than was anthropogenic Cu, perhaps because Pb was derived from the decades-long burning of leaded fuel. Sources for Cu were more limited and site-specific. Consequently, Cu has been cycling through the watershed longer than has Pb. Most Cu was conveyed through the drainage basin before 2000, thus the change in Cu concentration between 2000 and 2007 was minimal. In contrast to the concentrations of Cu and Pb, the concentration of Zn did not change significantly over the time period (Table 1) and the downstream variability of Zn changed little from 2000 to 2007 (Fig. 2; Table 2). Together, these findings suggest that upstream sources of Zn continued to supply the metal to the Lahn River between 2000 and 2007.

Reductions in upstream releases of heavy metals to watersheds have been shown to result in lower concentrations of heavy metals on downstream floodplains and in alluvial fills (Harland et al., 2000; Ciszewski, 2003; Grosbois et al., 2006; Bábek et al., 2008), although Outridge et al. (2011) noted a lag of several decades between the decline in metal emissions and a reduction in metal concentrations in downstream lake sediments at a contaminated site in Manitoba, Canada. When deposited on floodplain soils, uncontaminated or less contaminated sediments reduce surface metal concentrations (Leece et al., 2008). An influx of uniformly cleaner sediments to the Lahn does not adequately explain the changes documented here because only the concentrations of Cu and Pb decreased over the study period; the addition of uniformly less-contaminated sediments to the watershed would have been expected to reduce the concentrations of all three metals between 2000 and 2007. Rather, the varying changes in downstream trends and near-channel concentrations of the three metals between 2000 and 2007 suggest differences in the upstream sources of Cu, Pb, and Zn. Specifically, the source for Pb decreased most dramatically, the source for Cu less dramatically, and the source for Zn remained unchanged. Sediments deposited post-2000 still contained concentrations of Pb, Cu, and Zn that were above the background concentrations, but the sediments were less contaminated with Cu and Pb in 2007 than they had been in 2000.

Although Cu and Pb concentrations are declining at near-channel locations along the Lahn River, Martin (2009) reported an increase in Pb and Zn concentrations and no statistical change in Cu concentrations between 1995 and 2007 along five cross-valley transects on the Lahn River floodplain between Marburg and Wetzlar. That is, the reduction in Pb concentrations noted here at near-channel locations is not found at distal floodplain positions. Metal concentrations generally decrease with increasing distance from the Lahn River channel, evidence that metals are deposited primarily by fluvial processes (Martin, 2004). A possible explanation for the different temporal trends in metal concentrations between the present study and Martin (2009) is that proximal floodplain positions received inputs of alluvium containing lower Cu and Pb concentrations post-2000; whereas more distal parts of the floodplain, which are inundated less frequently, have not yet received measureable quantities of the cleaner sediments. Metal concentrations

Table 4

Enrichment factors for metals; enrichment factor calculated by dividing mean metal concentration by the mean background metal concentration (location abbreviations provided in Table 2).

	Cu	Pb	Zn
2000			
Entire Study Reach (Mb–Wb)	2.0	2.3	2.9
River section			
(Mb–Gi)	1.6	2.0	2.3
(Gi–Wz)	2.1	2.3	2.9
(Wz–Wb)	2.7	3.4	4.2
2007			
Entire study reach (Mb–Wb)	1.8	1.8	2.9
River section			
(Mb–Gi)	1.5	1.5	2.3
(Gi–Wz)	1.9	2.0	3.2
(Wz–Wb)	2.3	2.2	3.5

in surface soils at channel banks decreased more rapidly than did the concentrations in soils on more distal positions of the floodplain.

In 2000, the concentrations of Cu, Pb, and Zn were not related to the fine sediment content of surface soils (Table 3). Organic matter had slightly greater effect on metal concentrations (especially the Zn concentration), although the statistical relationship, even when significant at the 0.05 level, was minimal ($r^2 = 0.15$). In 2007, the relationship between clay content and metal concentrations was statistically significant, suggesting that metals were being transported with the fine sediment load, something noted in previous studies (e.g., Horowitz, 1985; Bubb and Lester, 1996). The strengthening of the influence of percent clay on metal content from 2000 to 2007 suggests that the physical process of sedimentation, which controls the dispersal of suspended sediments across the floodplain, has become increasingly important in explaining the spatial pattern of metal storage along the Lahn River.

6. Conclusion

Concentrations of Cu and Pb at near-channel positions along the Lahn River, central Germany, decreased between 2000 and 2007; in contrast, the concentration of Zn was statistically unchanged during the two periods. The Pb concentration decreased at 82% of the 33 sample sites; Cu concentrations declined at 70% of the sample sites. In contrast, Zn concentrations increased at 48% of the sample sites and decreased at 48%. When the study reach is divided into three river segments, Pb concentrations at near-channel sites declined in all three sections between 2000 and 2007, Cu concentrations declined at two of three sections, and Zn concentrations decreased at one of the three sections. The change in enrichment factors demonstrates that anthropogenic Pb contamination decreased more between 2000 and 2007 than did the contamination of Cu. In contrast, the level of Zn contamination for the entire study reach was unchanged between 2000 and 2007, but did decrease along the Wetzlar to Weilburg river segment.

The decrease in Pb concentrations adjacent to the channel indicates that the conveyance of Pb-contaminated sediments has diminished since 2000. The deposition of cleaner sediments along the channel suggests a reduction in the load of Pb-contaminated sediments conveyed to the study reach post-2000. The widespread release of Pb to the environment during decades of leaded fuel consumption resulted in the metal being spread over geographically broad areas of the Lahn River watershed. In contrast, Cu was likely released to the environment from point-specific locations (Scherer et al., 2003) and had a more direct pathway to surface water systems. Anthropogenic Cu began cycling through the watershed earlier than did Pb, and much of the Cu had been conveyed through the basin before the 2000 to 2007 study period. As a result, the Cu concentration decreased less than did the Pb concentration between 2000 and 2007. The concentration of Zn was unchanged between 2000 and 2007, an indication that the upstream source for the metal changed little during the study period. The storage of metals in upstream catchment soils has been termed “saturation” and produces a lag between the decline in metal emissions to the environment and the decrease in metal concentrations in downstream depositional environments (Nriagu et al., 1998; Outridge et al., 2011).

The decrease in Pb, and to a lesser extent Cu, concentrations appears to be restricted to near-channel locations. Along five cross-valley transects, Martin (2009) demonstrated that Pb and Zn concentrations increased, whereas Cu concentrations were unchanged, at distal parts of the floodplain between 1995 and 2007. Cleaner sediments are being deposited at near-channel locations, which are flooded more frequently, but are not reaching more distal parts of the floodplain. Presumably, sufficiently large floods will eventually spread cleaner sediments across the floodplain, reducing metal concentrations in surface soils of distal floodplain locations.

Recently deposited fluvial sediments adjacent to the Lahn River contain less Cu and Pb, ostensibly a reflection of several decades of controls

on heavy metals emissions in Germany. Higher concentrations of heavy metals remain in more deeply buried sediments and on distal parts of the floodplain (Martin, 2004, 2009), but near-channel surface soils are becoming cleaner. The presence of lower metal concentrations along the Lahn River would appear to confirm that environmental controls implemented over several decades are having the desired effect. The reduced concentrations of metals such as Cu and Pb in geomorphically active channel bank positions translate to a diminished risk of future downstream heavy metal contamination should these sediments be eroded and conveyed downriver.

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